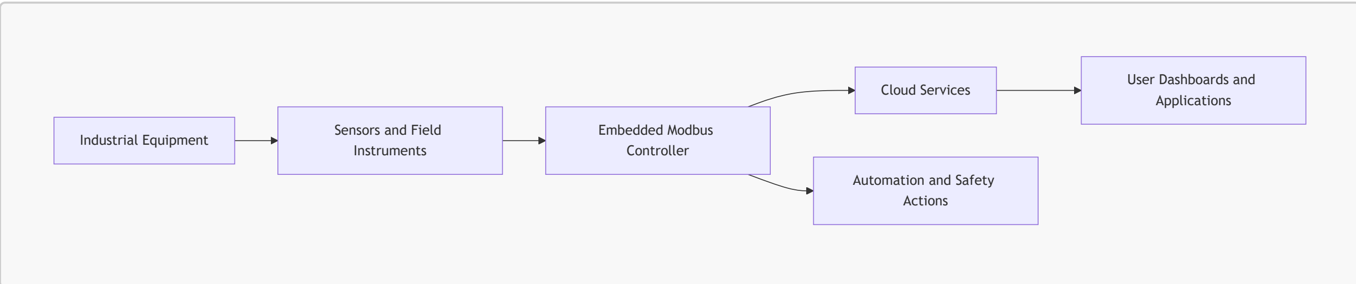
Smart Industrial IoT Monitoring & Control System Using Modbus Controller

1. Executive Summary

The Smart Industrial IoT Monitoring & Control System (SIIMCS) is a cloud-connected industrial monitoring and control solution.

SIIMCS uses an embedded controller to collect operating data from industrial sensors and field instruments through a Modbus communication interface. The system supports real-time monitoring, threshold-based alerts, rule-based control, cloud reporting, and maintenance planning.

The objective of this BRD version is to keep the requirements short, clear, reviewable, and executable for stakeholders.



2. Business Objectives

Objective	Description
Real-Time Monitoring	Collect and present current operating data from industrial equipment.
Rule-Based Control	Trigger predefined actions when operating conditions cross approved limits.
Alerts and Response	Notify responsible teams when sensor values require attention.
Integration	Share data with cloud platforms, dashboards, APIs, and enterprise systems.
Reliability	Continue safe monitoring during sensor, controller, or connectivity issues.
Maintenance Support	Use historical readings, operating history, and equipment condition changes to plan maintenance and reduce unplanned downtime.

3. Business Scope

3.1 In Scope

- Controller-based polling of three configured sensors or field instruments.

- Collection of temperature, vibration, gas, pressure, or equivalent industrial readings.
- Cloud-based storage and monitoring.
- User-facing dashboards and operational interfaces.
- Alert and escalation workflows.
- Rule-based automation for safety and efficiency.
- REST API integration with third-party systems.
- Local data storage on the controller during internet outage.
- Synchronization of locally stored data after connectivity returns.
- Sensor and controller health visibility.

3.2 Out of Scope

- Manufacturing physical sensors.
- Full SCADA replacement.
- On-premises-only deployment in the initial phase.
- Custom AI model training beyond predefined industrial parameters.
- Manual calibration workflow design.
- Custom hardware design for every industrial machine.

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4. Stakeholders

Stakeholder	**Responsibility**
Client Organization	Owns business goals, priorities, and acceptance decisions.
IT and Engineering Teams	Implement controller, cloud, interfaces, and integrations.
Operations Managers	Monitor equipment status, alerts, and response actions.
Maintenance Teams	Use equipment condition data to plan service activity.
Safety Teams	Define critical limits and emergency actions.
Third-Party Integrators	Connect SIIMCS with external enterprise systems.
QA and Validation Teams	Validate workflows, alerts, controls, and reliability scenarios.

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5. System Overview

SIIMCS consists of:

- Industrial equipment being monitored.
- Sensors or field instruments connected to equipment.
- Embedded controller that reads sensor data using Modbus.
- Cloud services for storage, analytics, alerts, and reporting.
- User-facing dashboards and applications.
- REST APIs for enterprise integration.
- Actuators for approved automation and safety actions.

The controller acts as the Modbus client. Connected devices expose readings through their Modbus interface.

6. Functional Requirements

6.1 Sensor Data Collection and Monitoring

ID	Requirement
BR-SEN-001	The controller shall collect data from three configured sensors or field instruments.
BR-SEN-002	The controller shall poll each configured device at a defined interval.
BR-SEN-003	The system shall display the latest available readings to users.
BR-SEN-004	The system shall maintain current health status for each sensor and for the controller.
BR-SEN-005	Polling frequency shall be configurable within approved operating limits.

6.2 Supported Device Types

The initial release shall support three active sensor positions per controller deployment.

Device Type	Business Purpose
Temperature Sensor	Monitor heat levels in industrial machinery.
Vibration Sensor	Detect abnormal machine behavior and wear conditions.
Gas Sensor	Detect hazardous gas conditions.
Pressure Sensor	Monitor process pressure in pipes, tanks, or machines.

6.3 Communication and Polling

ID	Requirement
BR-COM-001	The controller shall initiate every Modbus transaction.
BR-COM-002	The controller shall continuously poll Sensor-1, Sensor-2, and Sensor-3 at configurable intervals.
BR-COM-003	Polling shall be deterministic and shall not allow one sensor to block the others.
BR-COM-004	The controller shall continue polling healthy sensors even if one sensor fails.
BR-COM-005	Communication failures shall update sensor health and availability status.

6.4 Startup and Configuration

ID	Requirement
BR-CFG-001	The controller shall validate configuration during startup.
BR-CFG-002	Duplicate addresses, invalid ranges, unsupported functions, or malformed configuration shall be rejected.
BR-CFG-003	Polling shall start only after configuration validation is successful.
BR-CFG-004	Authorized users shall be able to update thresholds and polling settings.
BR-CFG-005	If a runtime change fails validation, the last valid configuration shall remain active.

6.5 Request and Response Validation

ID	Requirement
BR-VAL-001	The controller shall use a sensor reading only when it matches a valid controller request.
BR-VAL-002	The controller shall validate device identity before accepting data.
BR-VAL-003	Invalid, delayed, duplicate, or mismatched responses shall not update business views or trigger actions.
BR-VAL-004	Communication errors shall update data quality and sensor health.

6.6 Offline Storage and Recovery

ID	Requirement
BR-OFF-001	If internet connectivity is lost, the controller shall store collected data locally.
BR-OFF-002	When connectivity returns, the controller shall synchronize stored data with the cloud.
BR-OFF-003	The local retention window shall be configurable, such as up to 48 hours.
BR-OFF-004	The application shall show cloud synchronization status.

6.7 Alerts and Notifications

ID	Requirement
BR-ALT-001	Users shall be able to configure warning and critical thresholds.

ID	Requirement
BR-ALT-002	Alerts shall be available through operational interfaces and configurable notification channels.
BR-ALT-003	Event-driven alerts shall be supported for abnormal equipment conditions.
BR-ALT-004	Escalation shall occur if an alert is not acknowledged within configured time limits.
BR-ALT-005	Alerts shall include equipment, sensor type, value, severity, time, and recommended action.

6.8 Rule-Based Automation

The system shall support approved automation rules based on equipment condition.

Condition	Safety Level	Expected Action
Temperature exceeds threshold	Medium	Increase fan speed.
Vibration exceeds safety limit	Medium	Reduce motor speed.
Gas leakage detected	High	Trigger emergency shutdown.
Temperature warning and vibration high	Medium	Trigger warning and reduce machine speed by 20%.
Pressure low and temperature critical	High	Emergency shutdown and notify operators.

6.9 Sensor and Actuator Data Sheet

Type	Normal Range	Min Threshold	Max Threshold	Critical Level
Temperature Sensor	10-50°C	5-10°C	50-70°C	>70°C
Vibration Sensor	1-5 mm/s	0-1 mm/s	5-8 mm/s	>8 mm/s
Gas Sensor	10-12 psi	5-10 psi	12-13 psi	>13 psi
Pressure Sensor	30-50 psi	10-30 psi	50-60 psi	<10 psi or >60 psi
Fan Actuator	Variable Speed	Low Speed	High Speed	Max Speed
Motor Controller	0-100% Speed	10% Speed	90% Speed	Full Stop
Valve Controller	Open/Close	10% Open	90% Open	Emergency Shut
Shutdown Relay	Active/Inactive	Warning Level	Critical Level	Full Shutdown

6.10 Health and Data Status

ID	Requirement
BR-HLT-001	Sensor health shall be shown as Online, Degraded, Offline, or Protocol Error.
BR-HLT-002	One failed sensor shall not stop monitoring of the remaining sensors.
BR-DQ-001	Data shall be identified as Good, Stale, Invalid, or Unavailable.
BR-DQ-002	Latest good value shall be retained during temporary failure.
BR-DQ-003	Stale data shall be clearly identified in dashboards and APIs.

6.11 User Experience and Integration

ID	Requirement
BR-APP-001	The system shall display current readings, health, alerts, and recent status.
BR-APP-002	The system shall expose sensor data and status through REST APIs.
BR-APP-003	Third-party systems shall be able to integrate through approved secure interfaces.

6.12 Security

ID	Requirement
BR-SEC-001	Only authorized users shall configure sensors, thresholds, and automation rules.
BR-SEC-002	Access to APIs and control actions shall be authenticated and authorized.
BR-SEC-003	Network access to controller and cloud services shall follow enterprise security policy.
BR-SEC-004	Safety-critical actions shall require approved business rules.

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7. Non-Functional Requirements

Area	Requirement
Reliability	Monitoring shall continue safely when one sensor or one communication path fails.
Availability	Latest good data shall remain available during temporary disruptions.
Performance	Polling, alerting, and synchronization shall meet agreed business service levels.
Scalability	The design shall support future expansion to more assets and devices.
Maintainability	Thresholds, rules, and polling settings shall be configurable.
Security	Access to data and control actions shall be protected.
Embedded Safety	The controller shall operate within approved CPU, memory, storage, and watchdog limits.

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8. Acceptance Criteria

ID	Acceptance Criteria
AC-001	The controller polls three configured sensors successfully.
AC-002	Current readings, health, and alerts are visible to users.
AC-003	One sensor failure does not stop the remaining sensors from being monitored.
AC-004	Warning and critical alerts trigger correctly.
AC-005	Approved automation rules trigger the correct control actions.
AC-006	Data is stored locally during internet outage and synchronized after recovery.
AC-007	Runtime configuration updates are validated before activation.
AC-008	Security and access controls are enforced.
AC-009	Long-duration monitoring runs without controller crash or data loss.

9. Definition of Done

The feature is complete when:

- Three configured sensors are monitored successfully.
- Dashboards, alerts, APIs, and automation rules are working.
- Offline storage and data synchronization are verified.
- Sensor, controller, and network failure scenarios are tested.
- Security and access controls are verified.
- Stakeholders approve the business requirements and acceptance criteria.